

EXPERIMENT 23:

23. Construct a C program to implement first fit algorithm of memory management.

PSEUDO CODE :

Start

Input memory blocks

Input process sizes

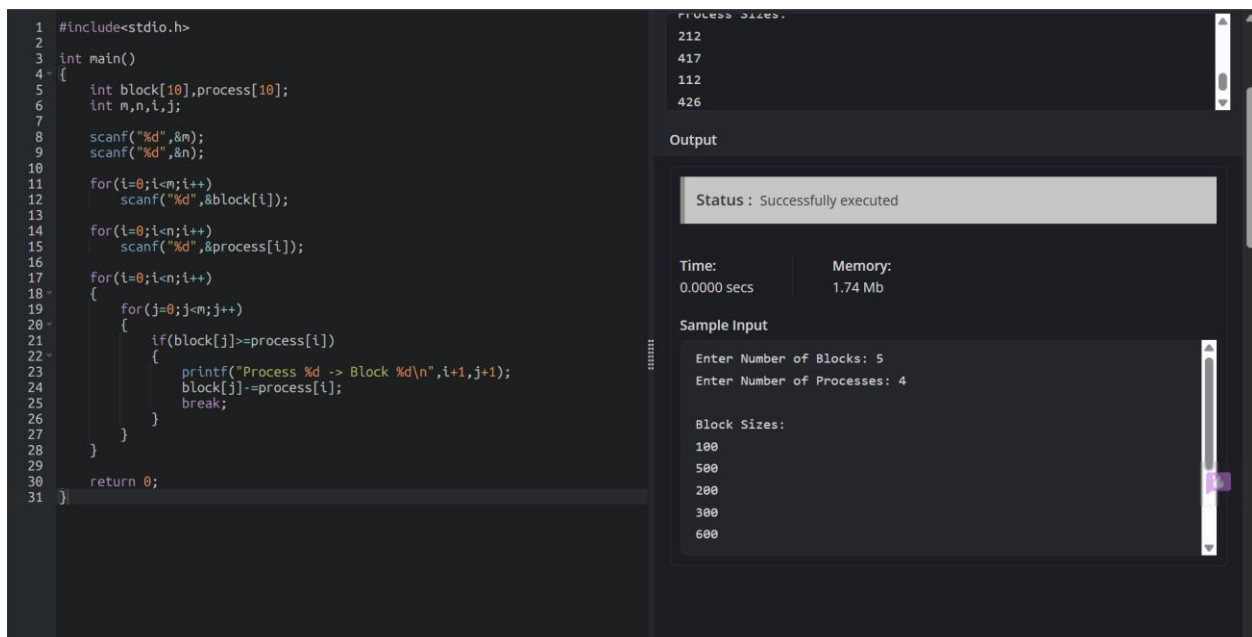
For each process

Find first suitable block

Allocate process

Display allocation

Stop



```
1 #include<stdio.h>
2
3 int main()
4 {
5     int block[10],process[10];
6     int m,n,i,j;
7
8     scanf("%d",&m);
9     scanf("%d",&n);
10
11     for(i=0;i<m;i++)
12         scanf("%d",&block[i]);
13
14     for(i=0;i<n;i++)
15         scanf("%d",&process[i]);
16
17     for(i=0;i<n;i++)
18     {
19         for(j=0;j<m;j++)
20         {
21             if(block[j]>=process[i])
22             {
23                 printf("Process %d -> Block %d\n",i+1,j+1);
24                 block[j]-=process[i];
25                 break;
26             }
27         }
28     }
29
30     return 0;
31 }
```

PROCESS SIZES:

212
417
112
426

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.74 Mb

Sample Input

Enter Number of Blocks: 5
Enter Number of Processes: 4

Block Sizes:

100
500
200
300
600

CODE:

```
#include<stdio.h>

int main()

{
    int
    block[10],process[10];

    int m,n,i,j;

    scanf("%d",&m);

    scanf("%d",&n);

    for(i=0;i<m;i++)

    scanf("%d",&block[i]);

    for(i=0;i<n;i++)

    scanf("%d",&process[i])
    ;

    for(i=0;i<n;i++)

    {

        for(j=0;j<m;j++)

        {
```

```
if(block[j]>=process[i])
```

```
{
```

```
printf("Process %d ->  
Block %d\n",i+1,j+1);
```

```
    block[j]-  
=process[i];
```

```
    break;
```

```
}
```

```
}
```

```
}
```

```
return 0;
```

```
}
```